VECTOR SPACES · WORKED CASES

The four subspaces, by matrix type

Three matrices with the same four subspaces computed for each. The point is the pattern across them — how much of each space collapses as the rank falls.

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CASES

TYPE	CONDITION	COL(A)	ROW(A)	NULL(A)	NULL(A^T)
NOTHING COLLAPSES					
 **Invertible square**	$r = n = m$	all of $\mathbb{R}^n$	all of $\mathbb{R}^n$	$\{\mathbf{0}\}$	$\{\mathbf{0}\}$
ALMOST EVERYTHING COLLAPSES					
 **Rank one**	$A = \mathbf{u}\mathbf{v}^\mathsf{T}$, $r = 1$	the line through $\mathbf{u}$	the line through $\mathbf{v}$	hyperplane, $\dim = n - 1$	hyperplane, $\dim = m - 1$
COLLAPSE SPLIT EVENLY					
 **Symmetric projection**	$A^2 = A$, $A = A^\mathsf{T}$	the target subspace S	the same as $\mathrm{Col}(A)$	$S^\perp$	the same as $\mathrm{Null}(A)$

Rank is the only variable. At $r = n$ nothing is lost and both null spaces are trivial; at $r = 1$ almost everything is lost and the null spaces are hyperplanes. Every row obeys $r + \text{nullity} = n$ on the domain side and $r + \text{left nullity} = m$ on the codomain side — the dimension accounting with three sets of numbers substituted in.